

Backup Infrastructure — Target Design v3

& Build Plan

Full-capability design leveraging Veeam Enterprise Plus · KOR-driven · supersedes v1/v2 (2026-07-23)

Date: 2026-07-23 **Owner:** Ian Lalonde (KOR) **Legend:** KOR T-NET TOUCHPOINT DONE

Objective. Not merely "backups that work," but a system that **proves it works, defends itself, recovers fast, keeps clean-room history, and copies off-site** — self-verifying and largely hands-off, using features already inside the Enterprise Plus licence. KOR leads; T-Net's licence + Cloud Connect are retained inputs, kept informed rather than gating.

1. The five pillars of "immaculate"

① Verified recovery — the system proves itself

- **SureBackup** (Enterprise Plus): on a schedule, boots each backup in an isolated virtual lab (fenced from production), verifies OS boot + services + optional app tests, tears down, and emails a pass/fail. Automated recovery testing — replaces manual restore drills.
- **Health Checks:** periodic auto re-read of backup files to detect storage corruption early (the failure that silently killed the FS007 backup years ago).

② Ransomware resilience — layered

- **Veeam Threat Hunter** (installed on APP01 with 12.3): inline malware/ransomware signature scanning of backups as they're created; infected points flagged so you never restore the infection.
- **Suspicious-activity detection:** alerts on anomalies in the backup data (encryption/deletion spikes) — an early tripwire.
- **Immutable archives:** point-in-time copies on object storage with object-lock (or a hardened Linux repo) — cannot be altered or deleted for their retention, even if the whole environment incl. the Veeam server is compromised.

③ Fast recovery (RTO)

- **File-level restore** incl. deduplicated volumes — **working** (APP01 mount server).
- **Instant VM Recovery:** boot a dead VM from backup in minutes; validated via the SureBackup lab. Add a cache disk to APP01 if FS01 IVMR is wanted (its C: is ~43 GB).
- *Option:* **Cloud Connect Replication** (standby VMs at T-Net) for near-zero-RTO failover — heavier; adopt only if RTO warrants.

④ Clean-room history — automated point-in-time

- **GFS retention:** auto-keep weekly/monthly/semi-annual fulls by policy on the archive tier — point-in-time copies with zero human discipline. **Export Backup** stays for ad-hoc "grab a full before a big change."
- Protects against slow intrusions/corruption that would poison all recent backups.

⑤ Efficient, resilient storage & off-site

- **Forward incremental + weekly synthetic full** (measured ~4× faster, ~1/3 the space of reverse here); **ReFS/XFS fast-clone** for cheap synthetic fulls; **BitLocker** skips swap/deleted blocks.
- **Backup Copy** → **Cloud Connect** off-site (+ WAN accel if T-Net offers it). 3-2-1: primary (Synology), off-site (Cloud), archive (E: → immutable).

2. Target architecture

Production (vSphere / UC3200) → Veeam BK01 (12.3.2)

- ① **Primary** → **Synology (F:)** forward-incr + synthetic full · mount server APP01 · Health Check + Threat Hunter on
- ② **Off-site** → **Cloud Connect (T-Net)** Backup Copy over the wire (KOR-built)
- ③ **Archive** → **E: (repurposed)** GFS + Export point-in-time fulls · immutable target in phase 2
- ④ **Verification** → **SureBackup virtual lab** boots & tests backups on schedule, emails pass/fail

3. Build plan — ordered, KOR-driven

Phase A — Off-site copy of FS01 (the unlock) KOR OPTIONAL SPEED-UP

1. Home → **Backup Copy Job** → source `Kor-FS01` → target `Kor-Replication` (Cloud Connect); compression on; throttle for business hours, full off-hours.
2. Start; ~5.8 TB seeds over ~a week (resumable, 182 Mbps). Optionally ask T-Net re: target WAN accelerator / seed-drive import.
3. Verify FS01 under Backups → Cloud with current points. **Off-site gap closed.**

Phase B — Verified recovery: SureBackup + Health Checks KOR

1. Enable **Health Check** on all backup jobs (Edit job → Storage → Advanced → Maintenance → periodic health check).
2. Build a **Virtual Lab** on a host with headroom (Backup Infrastructure → SureBackup → Virtual Labs) — an isolated, fenced network.
3. Create an **Application Group** (the VMs to verify) and a **SureBackup job** — schedule it after nightly backups; enable boot + ping tests. Result emails to KOR.
4. From here, recovery is proven automatically — the manual restore-test cadence becomes the fallback, not the primary assurance.

Phase C — Ransomware hardening KOR OPTION

1. Enable **Threat Hunter / inline malware scanning** (Security & Compliance in the 12.3 ribbon) so backups are scanned as written.
2. Turn on **suspicious-activity / anomaly alerts**; route to KOR email.
3. Plan an **immutable archive target** — object storage with object-lock (Wasabi/B2 capacity tier) or a hardened Linux repo — for the GFS/Export archives (Phase E storage).

Phase D — Reclaim E: & stand up the archive tier KOR

1. Confirm restores (Synology + new Cloud copy) pass — SureBackup already doing this.
2. Delete the **old FS01 reverse-incr chain on E:** (Backups → Disk → Delete from disk) → reclaims ~23 TB. E: becomes the archive tier.
3. **Move ManualNew to E:** (right-click → Move backup) → frees ~12.7 TB on the Synology; rename job `Archive-2026-03`. This is archive #1.

Phase E — Automated archives, optimisation, DR options KOR

- 1. Enable **GFS** on the FS01 job (keep e.g. semi-annual fulls, 2 yrs) landing on the archive/immutable tier — point-in-time, automated.
- 2. Convert `Kor-VMs-New` & `Vcenter` to **forward incremental + synthetic full**; enable ReFS/XFS fast-clone on the backup volume if reformatting is on the table.
- 3. Finish tidy-ups: retire APP003/APP004 (host .10, have backups); resolve `Kor-RDS01` / `_1` duplicate.
- 4. *Evaluate*: Cloud Connect **Replication** for near-zero-RTO DR of FS01 (only if the RTO justifies the extra footprint).

4. Capability coverage — Enterprise Plus, fully used

Capability	Feature	Status
Deduplicated file-level restore	APP01 (Server 2025) mount server	Done
Automated recovery verification	SureBackup + Health Checks	Phase B
Inline malware/ransomware scan	Veeam Threat Hunter	Phase C
Anomaly / suspicious-activity alerts	Security & Compliance (12.3)	Phase C
Immutable, ransomware-proof archives	Object-lock storage / hardened repo	Phase C/E
Instant recovery (RTO)	Instant VM Recovery (+ APP01 cache)	Phase B/E
Point-in-time / clean-room history	GFS retention + Export Backup	Phase D/E
Off-site copy	Backup Copy → Cloud Connect (+ WAN accel)	Phase A
Storage efficiency	Forward-incr + synthetic full + fast-clone + BitLocker	Phase E
Space reclaimed	~20 TB prod done; ~23 TB E: + ~12.7 TB Synology pending	In progress
Alerting	Email notifications (verified) — Veeam ONE optional	Done

5. T-Net — informed, and what they own

- Informed (courtesy, no sign-off):** Veeam 12.3.2 upgrade; APP01 mount server; ~20 TB retired VMs; the redesign incl. SureBackup, Threat Hunter, forward-incr, archive tier; FS01 added to off-site.
- Retained dependencies:** the Enterprise Plus **licence** (confirm coverage); the **Cloud Connect** endpoint (optional WAN accelerator + seed-drive; quota headroom); **monitoring/alerting SLA** and **vCenter certificate ownership** (renews 2028-07-14).

KOR Structural Operations, 2026-07-23. Supersedes v1/v2. Designed to use the Enterprise Plus feature set fully — SureBackup, Health Checks, Threat Hunter, GFS, immutability, Instant Recovery — most of which require no additional licensing. Seed estimates based on a measured 182 Mbps upload. Shared with T-Net as the redesign record.